POWER PULSE

Protection Basics Important Formulae

Notes by Balaji Sir

1 Requirements

1. Selectivity, speed, sensitivity, reliability (dependability + security), stability
2. $$R_{system} = 1 - (1 - R_{primary})(1 - R_{backup})$$

independent primary and backup

2 Current transformer

3. $$K_n = \frac{I_{pn}}{I_{sn}}, \qquad \text{Ratio error} = \frac{K_n I_s - I_p}{I_p} \times 100$$
4. $$\text{Burden (VA)} = I_{sn}^2 \, Z_b, \qquad R_b = \frac{VA}{I_{sn}^2}$$
5. $$V_k \approx ALF \times I_{sn} \, (R_{ct} + R_b) \qquad (\text{5P20: } 5\% \text{ composite error at } 20 \times I_n)$$
6. CT secondary must never be open-circuited; VT secondary must never be short-circuited

3 Differential protection

7. $$I_d = |I_1 - I_2|, \qquad \text{operate if } I_d > k \, \frac{I_1 + I_2}{2} + I_{min}$$

percentage (biased) differential

8. $$\text{Transformer: CT ratio } = \frac{I_{rated}}{1 \text{ or } 5 \text{ A}}; \quad \text{Y–}\Delta\text{: CTs in } \Delta \text{ on the Y side, Y on the } \Delta \text{ side}$$
9. $$I_{rated} = \frac{S}{\sqrt{3} \, V_L}$$
10. $$\text{Inrush: 2nd harmonic } > 15\text{–}20\% \Rightarrow \text{restrain}$$
11. $$\text{Unprotected winding (REF)} = \frac{I_{setting}}{I_{f,max}} \times 100 \ \%$$